

Lecture 12

Theory of General Linear System II: Independence, Null space and Span

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Part 0 Review and Outline

3 mins

Recall

In the last lecture ...

- Definition of Linear space
- Definition of subspace:

Formal Definition: Linear Space

Definition 11.1 (Linear space)

First time in your study:
Define a subject by “structure”!

Suppose V is a set associated with two operations:

- (i) Addition “+”: $\mathbf{u} + \mathbf{v} \in V, \forall \mathbf{u} \in V, \mathbf{v} \in V$.
- (ii) Scalar multiplication: $\alpha \mathbf{u} \in V, \forall \alpha \in \mathbb{R}, \mathbf{u} \in V$.

Closed under LC

V is called a linear space over $\mathbb{R}$ if the 8 axioms hold:

(A1) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}, \forall \mathbf{u}, \mathbf{v} \in V$.

(A2) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + \mathbf{v} + \mathbf{w}, \forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in V$.

(A3) There exists a element $\mathbf{0}$ s.t. $\mathbf{u} + \mathbf{0} = \mathbf{u}, \forall \mathbf{u} \in V$.

(A4) If $\mathbf{u} \in V$, then there exists $-\mathbf{u} = (-1)\mathbf{u}$, s.t. $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.

(A5) $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}, \forall \alpha \in \mathbb{R}, \mathbf{u}, \mathbf{v} \in V$.

(A6) $(\alpha + \beta)\mathbf{u} = \alpha\mathbf{u} + \beta\mathbf{u}, \forall \alpha, \beta \in \mathbb{R}, \mathbf{u} \in V$.

(A7) $\alpha(\beta\mathbf{u}) = (\alpha\beta)\mathbf{u}, \forall \alpha, \beta \in \mathbb{R}, \mathbf{u} \in V$.

(A8) $1\mathbf{u} = \mathbf{u}$.

Rules

Key elements:
0, 1, -1

Remark: Linear space is also called “[vector space](#)”
(in textbook as well); we will use linear space more often.

Subspace

Definition 11.2 (subspace)

Suppose V is a linear space.

We say W is a subspace of V if two conditions hold:

- i) W is a subset of V ;
- ii) W is a linear space.

In words: A subspace of V is a subset that is itself a linear space.

Proposition 11.2 (criteria of subspace)

Suppose V is a linear space. W is a subspace of V if:

- i) W is a nonempty subset of V ;
- ii) W is closed under addition: $\mathbf{u} + \mathbf{v} \in W, \forall \mathbf{u}, \mathbf{v} \in W$.
- iii) W is closed under scalar multiplication: $\alpha \mathbf{u} \in W, \forall \mathbf{u} \in W, \alpha \in \mathbb{R}$.

In words: A subspace of V is a nonempty subset that is closed under linear combination.

Property: A subspace W must contain the zero element: $\mathbf{0} \in W$.

NOT necessarily
the real number 0.

Lec 12 Null Space, Column Space and Span

Outline: Independence, Null space and Span

1. Linear dependence
2. Null space
3. Span

Strang's book v5: Sec 3.1

Learning goals: After this lecture, you should be able to

1. Verify whether a collection of vectors is linearly independent or not
- 2 Tell the relation of the null space and linear system
- 3 Compute the span of a set; Explain why the span is a subspace

Part I Linear Independence

Strang's book Sec. 3.4

- Definition
- Relation to linear system

27 mins

Motivation: Example of Unique Solution

Rectangular system can have unique solution.

Example:

$$\begin{cases} x_1 + x_2 = 1 & \textcircled{1} \\ 2x_1 + 3x_2 = -1 & \textcircled{2} \\ 5x_1 + 7x_2 = -1 & \textcircled{3} \end{cases}$$

$$\textcircled{1} + 2 \cdot \textcircled{2} = \textcircled{3},$$

so equation $\textcircled{3}$ is *redundant*

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 3 & -1 \\ 5 & 7 & -1 \end{array} \right] \begin{matrix} v_1 \\ v_2 \\ v_3 \end{matrix}$$

$$\begin{aligned} & [1, 1, 1] && v_1 \\ & + 2 \cdot [2, 3, -1] && + 2v_2 \\ & = [5, 7, -1] && = v_3 \end{aligned}$$

Equivalent to the following system:

$$\begin{cases} x_1 + x_2 = 1 & \textcircled{1} \\ 2x_1 + 3x_2 = -1 & \textcircled{2} \end{cases}$$

Rewrite as LC of v_i 's: LC of v_i 's = 0

$$v_1 + 2v_2 + (-1)v_3 = 0$$

Linear Dependence and Independence

Definition (linear dependence)

线性相关

Suppose V is a linear space over $\mathbb{R}$. Suppose $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \in V$.

We say $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are **linearly dependent** if

$\exists \mathbf{c} = (c_1, c_2, \dots, c_k) \in \mathbb{R}^k, \mathbf{c} \neq \mathbf{0}$, such that

$$c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k = \mathbf{0}$$

Think: Can we remove condition $(c_1, \dots, c_k) \neq (0, 0, \dots, 0)$?

If $c_i = 0, \forall i$, then $\sum_i c_i u_i = 0$ for any u_i 's.

Linear Dependence and Independence

Definition (linear dependence)

Suppose V is a linear space over $\mathbb{R}$. Suppose $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \in V$.

We say $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are linearly dependent if

$$\exists \mathbf{c} = (c_1, c_2, \dots, c_k) \in \mathbb{R}^k, \mathbf{c} \neq \mathbf{0}, \text{ such that } c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k = 0.$$

Think: Can we remove condition $(c_1, \dots, c_k) \neq (0, 0, \dots, 0)$?

Definition (linear independence) *线性独立*

Suppose V is a linear space over $\mathbb{R}$. Suppose $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \in V$.

We say $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are linearly independent

If $\forall \mathbf{c} = (c_1, c_2, \dots, c_k) \in \mathbb{R}^k \setminus \{\mathbf{0}\}$, we have

$$c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k \neq 0.$$

Or: $c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k = 0$ only happens when $c_1 = \dots = c_k = 0$.

Remark: We will say the set $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is linearly independent (or dependent)
If the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are linearly independent (or dependent).

Examples

$-2 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} = 0$ are linearly dependant (D).

$\mathbf{u}, 2\mathbf{u}$ are linearly D

(-1) $\mathbf{u}, \mathbf{v}, \mathbf{u} + \mathbf{v} = 0$ are linearly D

$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ are linearly Independent (I)

Assume $\alpha_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = 0,$

then $\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ 0 \end{bmatrix} = 0 \Rightarrow \alpha_1 = \alpha_2 = \alpha_3 = 0.$
so independent

Informal:

Red, Green, Yellow are linearly dependent since Yellow = Red + Green

Exercise: D or I (Dependent or Independent)?

Suppose $\mathbf{u}, \mathbf{v}$ are non-zero size- n vectors and distinct.

$\mathbf{u}, \mathbf{v}, \mathbf{0}$ are linearly D

Need to find $(C_1, C_2, C_3) \neq (0, 0, 0)$ s.t.

$$C_1 \mathbf{u} + C_2 \mathbf{v} + C_3 \mathbf{0} = \mathbf{0}.$$

$$\left(\begin{array}{l} \Rightarrow C_1 \mathbf{u} = -C_2 \mathbf{v} \\ \Rightarrow \mathbf{u} = -\frac{C_2}{C_1} \mathbf{v} \end{array} \right)$$

Pick $C_1 = C_2 = 0$, $C_3 = 1$, it holds.

$\mathbf{u}, \underbrace{2\mathbf{u}}_{\text{all } x}, \underbrace{2\mathbf{v}}_{\text{all } y}, 2\mathbf{u} - 5\mathbf{v}$ are linearly D

find LC of x & $y = \mathbf{u}$.

$$\begin{aligned} 5x + 2y &= \underline{5}(2\mathbf{u} + \underline{2\mathbf{v}}) + \underline{2}(2\mathbf{u} - \underline{5\mathbf{v}}) = \underline{14}\mathbf{u}. \\ &= \underbrace{(5 \times 2 - 2 \times 5)}_0 \mathbf{v} + (5 \times 2 + 2 \times 2) \mathbf{u} \end{aligned}$$

$$\sim 5x + 2y - 14\mathbf{u} = 0.$$

Exercise: D or I (Dependent or Independent)?

Columns of $B = \begin{bmatrix} 1 & 0 & a & 0 & c \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ are linearly D.

$\underline{e_1} \quad \underline{e_2} \quad \underline{u} \quad \underline{e_3} \quad \underline{v}$

$$u = a \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = ae_1 + be_2, \text{ so } ae_1 + be_2 + (-1) \cdot u = 0$$

Columns of $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ are linearly I.

$\vec{a_1} \quad \vec{a_2} \quad \vec{a_3}$

If $c_1 \vec{a_1} + c_2 \vec{a_2} + c_3 \vec{a_3} = 0$, then $\begin{bmatrix} c_1 \\ -c_2 \\ c_3 \\ c_3 \end{bmatrix} = 0$, then $c_1 = c_2 = c_3 = 0$

The expression is LC of columns, which is a matrix-vector product.
A very useful observation leading to an interesting result next.

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matrix-vec product

$$A\vec{x} = x_1\vec{a}_1 + \dots + x_n\vec{a}_n$$

LC of columns.

$$A\vec{c} = c_1\vec{a}_1 + \dots + c_n\vec{a}_n$$

LC of columns

$$\vec{c}A.$$

Linear dependence of Column Vectors

Claim: Columns of matrix A are linearly dependent iff $Ax=0$ has nonzero solution.

Proof:

Let $A = [\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n]$.

$\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly dependent,

$$\Leftrightarrow \exists (c_1, \dots, c_n) \neq 0 \text{ s.t. } c_1 \vec{v}_1 + \dots + c_n \vec{v}_n = 0$$

$$\Leftrightarrow \exists \vec{c} \neq 0, \text{ s.t. } A\vec{c} = 0 \quad [\text{by column form of matrix-vec product}]$$

$$\Leftrightarrow Ax=0 \text{ has non zero solution}$$

Special Case: Square Matrix

Columns of matrix A are linearly dependent iff
 $Ax=0$ has nonzero solution.

This is true for any rectangular matrix.

For **square matrix**, we can say more.

^(for square matrix A)
Lec 11 says “ $Ax=0$ has nonzero solution” is equivalent to “NOT invertible”, thus

Corollary 12.1: Square matrix A is invertible iff
the columns are linearly independent.

Invertibility Conditions (to be continued)

Theorem (Equivalent Conditions for Invertibility)

Let $A \in \mathbb{R}^{n \times n}$. (square matrix)

The following statements are equivalent:

1. A is invertible
2. The linear system $A\mathbf{x} = \mathbf{0}$ has a unique solution $\mathbf{x} = \mathbf{0}$
3. A is a product of elementary matrices
4. A has n pivots.
5. The columns of A are linearly independent
6. ...
7. ...
8. ...

Equation solving

Checking Linear Independence

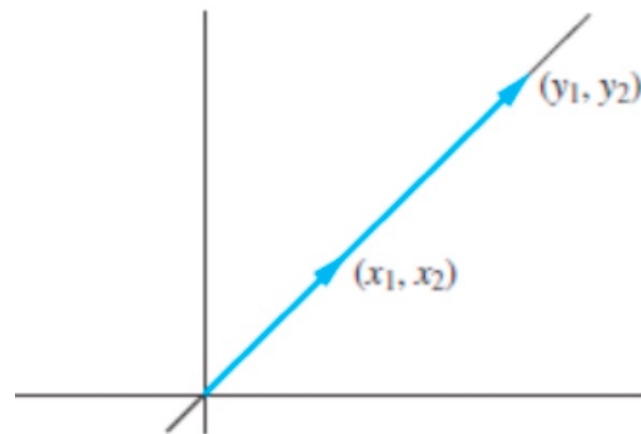
How to Check linear independence?

If the space is $\mathbb{R}^n$, then just need to solve a linear system!

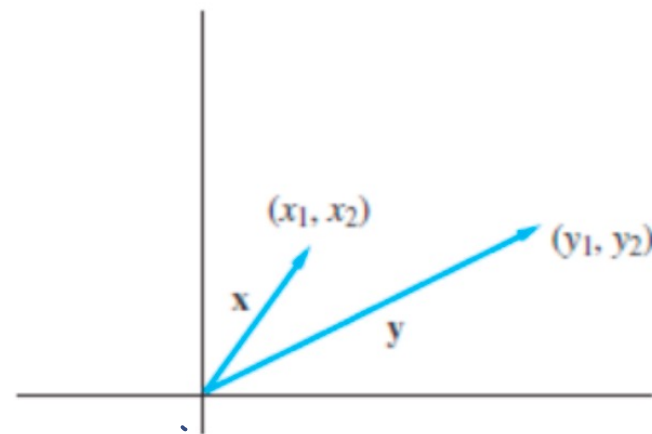
If the space is NOT $\mathbb{R}^n$, need extra tools
(not discussed here)

Geometry

In the plane:

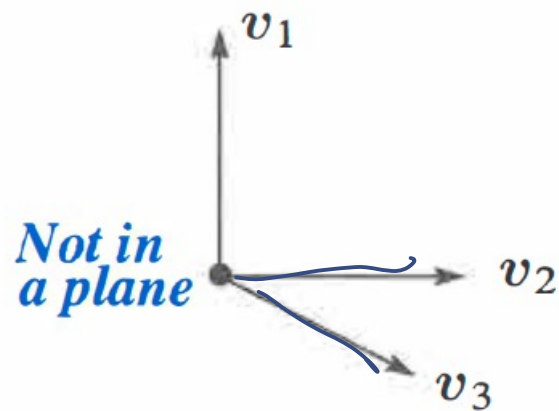


(a) x and y linearly dependent

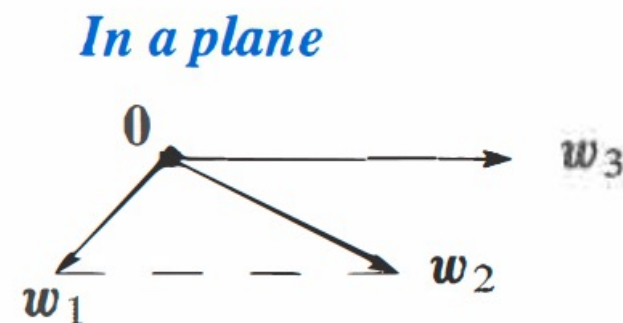


(b) x and y linearly independent

In a 3D space:



Linearly independent vectors



Linearly dependent vectors

Part II Null Space

—Null space, or Solution space of $Ax = 0$

10 mins

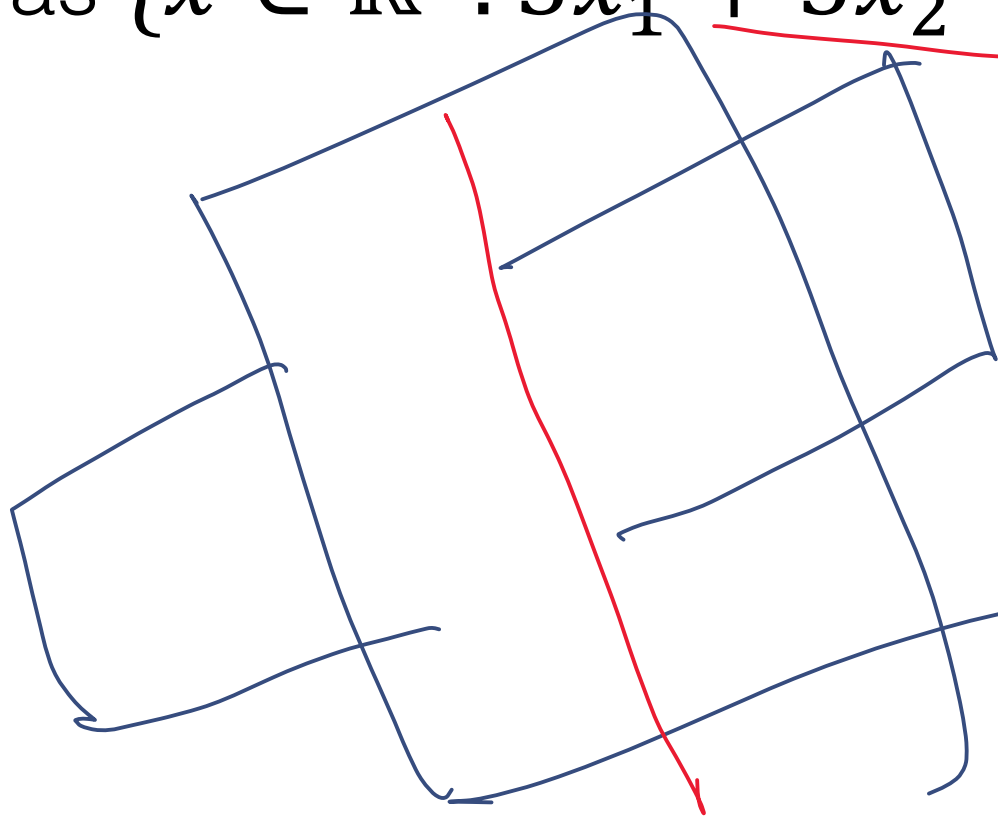
Taking intersection?

Eg $\{x \in \mathbb{R}^3 : 3x_1 + 5x_2 + x_3 = 0\}$ is a subspace of $\mathbb{R}^3$.

Eg $\{x \in \mathbb{R}^3 : 3x_1 = x_3\}$ is a subspace of $\mathbb{R}^3$.

What about the **intersection** of the two subspaces?

Expressed as $\{x \in \mathbb{R}^3 : 3x_1 + 5x_2 + x_3 = 0, 3x_1 - x_3 = 0\}$



Taking intersection?

Eg $\{x \in \mathbb{R}^3 : 3x_1 + 5x_2 + x_3 = 0\}$ is a subspace of $\mathbb{R}^3$.

Eg $\{x \in \mathbb{R}^3 : 3x_1 = x_3\}$ is a subspace of $\mathbb{R}^3$.

What about the **intersection** of the two subspaces?

Expressed as $\{x \in \mathbb{R}^3 : 3x_1 + 5x_2 + x_3 = 0, 3x_1 - x_3 = 0\}$

Still a subspace: intuitively, the intersection of two planes (passing 0) is a line (passing 0), thus a linear space.

One more step (pack into one matrix equation):

$$\left\{ x \in \mathbb{R}^3 : \begin{matrix} \boxed{\begin{matrix} 3 & 5 & 1 \\ 3 & 0 & -1 \end{matrix}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \boxed{\begin{matrix} 0 \\ 0 \end{matrix}} \end{matrix} \right\}.$$

$\cong A$

This is the solution set of $Ax=0$.

same set

Homogeneous Linear System

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Definition (homogeneous linear system)

A homogeneous linear system is

$$Ax = 0$$

where $A \in \mathbb{R}^{m \times n}$ are given and $x \in \mathbb{R}^{n \times 1}$ is the variable

In words: a homogeneous linear system is a linear system with RHS being **0**.

Homogeneous Linear System

Definition (homogeneous linear system)

A homogeneous linear system is

$$Ax = 0$$

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In words: a homogeneous linear system is a linear system with RHS being **0**.

Theorem 12.1

The solution set of a homogeneous linear system $Ax=0$ is a linear space.

Remark: Also a subspace of $\mathbb{R}^n$.

Definition 11.3 (null space)

The solution set of $Ax=0$ is called the null space of A , denoted as $N(A)$.

Proof

Break

Proof of Thm 12.1:

W is a subset of $\mathbb{R}^n$ (a linear space).

Denote the solution set as W . Need to verify:

(i) **W is non-empty:** W contains 0 since $A \mathbf{0} = \mathbf{0}$.
 $\Rightarrow \mathbf{0} \in W$

(ii) W is closed under scalar multiplication:

need to prove $\mathbf{x} \in W, \alpha \in \mathbb{R} \Rightarrow \alpha \mathbf{x} \in W$.

$$\begin{aligned} \mathbf{x} \in W &\stackrel{\text{def}}{\Rightarrow} A\mathbf{x} = \mathbf{0} \Rightarrow \alpha(A\mathbf{x}) = \mathbf{0} \cdot \alpha = \mathbf{0} \\ &\Rightarrow A(\alpha \mathbf{x}) = \mathbf{0} \Rightarrow \alpha \mathbf{x} \in W. \end{aligned}$$

(iii) W is closed under addition:

need to prove $\mathbf{x} \in W, \mathbf{y} \in W \Rightarrow \mathbf{x} + \mathbf{y} \in W$.

$$\begin{aligned} \left. \begin{array}{l} \mathbf{x} \in W \Rightarrow A\mathbf{x} = \mathbf{0} \\ \mathbf{y} \in W \Rightarrow A\mathbf{y} = \mathbf{0} \end{array} \right\} &\Rightarrow A\mathbf{x} + A\mathbf{y} = \mathbf{0} \\ &\Rightarrow A(\mathbf{x} + \mathbf{y}) = \mathbf{0} \Rightarrow \mathbf{x} + \mathbf{y} \in W. \end{aligned}$$

Think. $\left\{ \mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right\}$ is NOT a linear space.

Part III Span

—Sec. 3.1 of Strang's book v5

30 mins

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- Discussed $\rightarrow$ linear spaces $\mathbb{R}^n$, $\mathbb{R}^{n \times m}$, P_n
- null space of $Ax=0$

Next, a new class of linear spaces.

Two Ways to Generate Subspaces

Solution set of linear equation +
Taking **intersection**

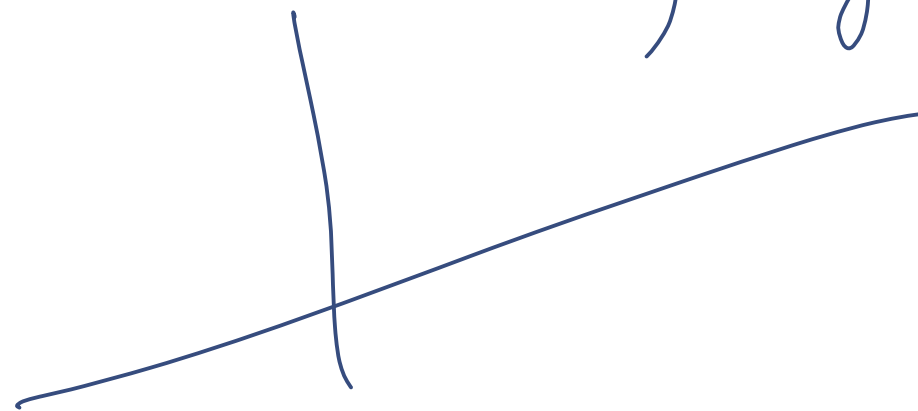
$\implies$ solution set of system of linear equations
(called null space)

getting smaller



Next, we show another important mechanism
of generating a subspace: **span** (linear combination).

getting bigger

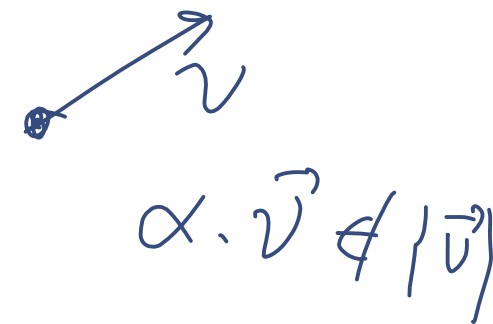


A Nontrivial Discrete Set is NOT Linear Space

Observation: A nontrivial discrete set is not a linear space.

Trivial discrete set: $\{\mathbf{0}\}$ is a linear space. here, 0 is the zero element of a linear space V .

Eg1: a single point $\{\mathbf{v}\}$ is NOT a linear space, for nonzero $\mathbf{v}$.



Eg2: a set of two points $\{\mathbf{u}, \mathbf{v}\}$ is NOT a linear space.

Motivating Question:

How to expand the set to a linear space?

Motivation: Expanding to Linear Space

Motivating Question:

How to expand a discrete set to a linear space?

Surely, the whole space that contains the set is a linear space.

That's NOT interesting.

An **interesting** question is:

Motivation: Expanding to Linear Space

Motivating Question:

How to expand a discrete set to a linear space?

Surely, the whole space that contains the set is a linear space.

That's NOT interesting.

An **interesting** question is:

What is the **minimal** linear space
that contains $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$?

Expanding One Element

Eg1: a single point $\{\mathbf{v}\}$ is NOT a linear space.

$$\alpha \vec{v} \notin \{\vec{v}\}.$$

Let's analyze what a minimal space V should contain.

$$2\vec{v} \notin \{\vec{v}\}, -0.5\vec{v} \notin \{\vec{v}\}.$$

First, $2\mathbf{v}$ should be in V ; $-0.5\mathbf{v}$ should be in V ; ...

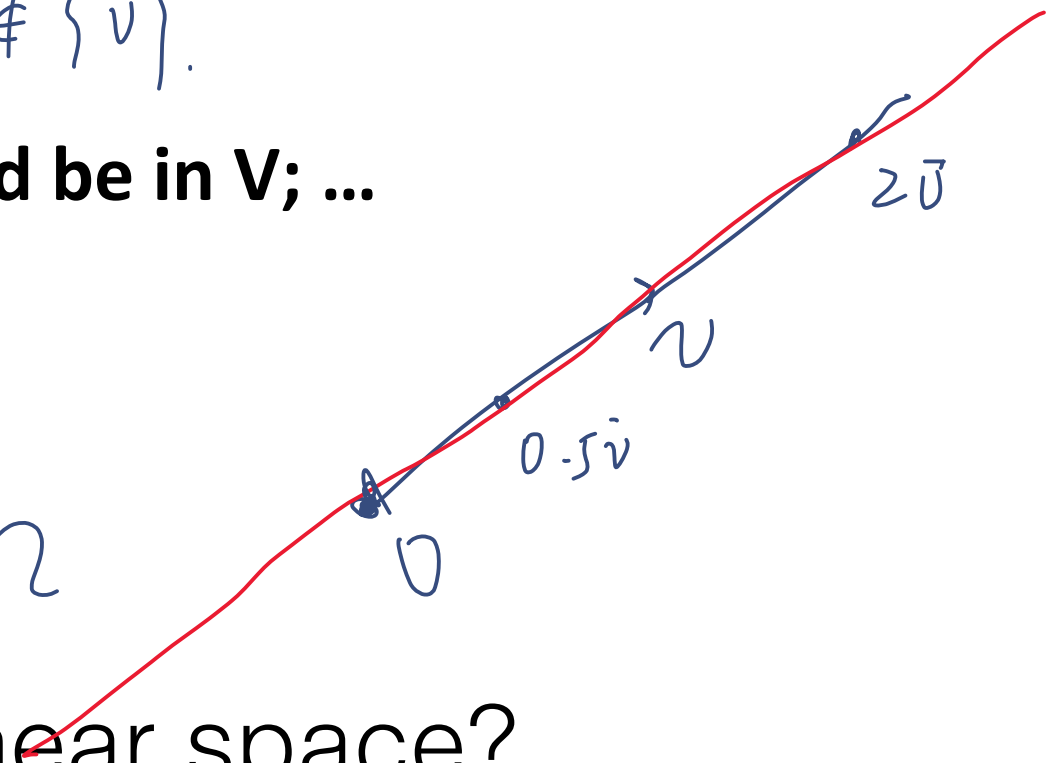
In short, $\alpha \vec{v}, \forall \alpha$ should be in V .

V is at least $\{\alpha \vec{v} : \alpha \in \mathbb{R}\} \triangleq \mathcal{L}$

Is this enough for it to be linear space?

Yes closed under scalar mult. If $\vec{u} \in \mathcal{L}, \beta \in \mathbb{R}$, then suppose $\vec{u} = \alpha \vec{v}$,
 $\beta \vec{u} = \beta(\alpha \vec{v}) = (\beta\alpha) \vec{v} \in \mathcal{L}.$

closed under addition. $\alpha \vec{v} + \beta \vec{v} = (\alpha + \beta) \vec{v} \in \mathcal{L}.$



Expanding Two Elements

Eg2: a set of two points $\{\mathbf{u}, \mathbf{v}\}$ is NOT a linear space.

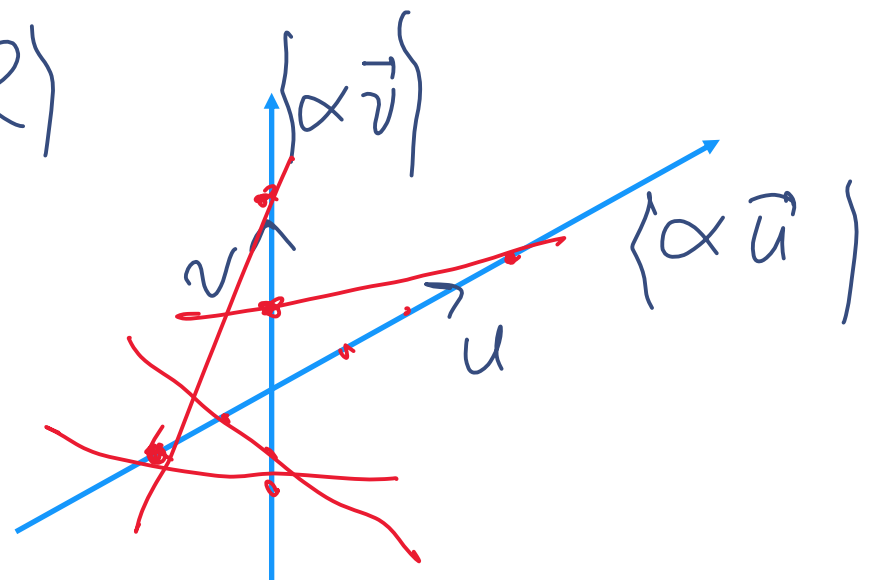
Let's analyze what a minimal space V should contain.

First, $\alpha\mathbf{u}$, $\alpha\mathbf{v}$ should be in V .

V is at least $\{\alpha\vec{u}, \alpha\vec{v} : \alpha \in \mathbb{R}\}$

Is this enough for it to be linear space?

No. Check “+”



Is the union of two lines a linear space?
No.

Expanding Two Elements

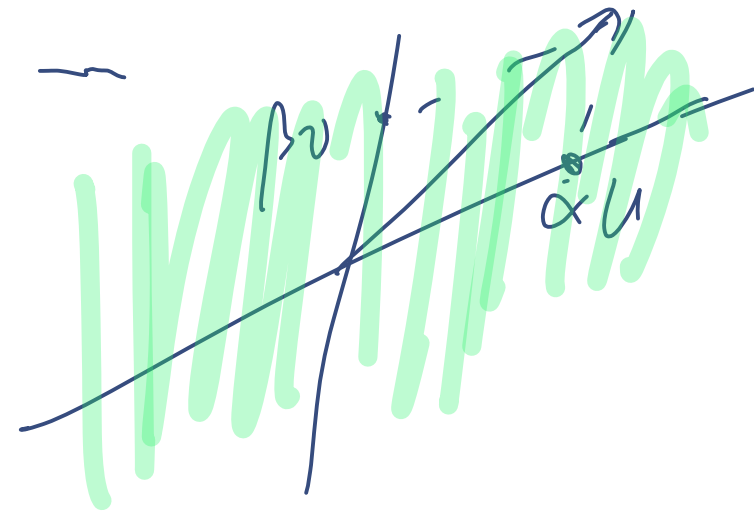
Eg2: a set of two points $\{\mathbf{u}, \mathbf{v}\}$ is NOT a linear space.

Let's analyze what a minimal space V should contain.

First, $\alpha\mathbf{u}, \alpha\mathbf{v}$ should be in V .

V is at least _____

Is this enough for it to be linear space?



Second, $\alpha\mathbf{u} + \beta\mathbf{v}$ should be in V . $M \triangleq \{\alpha\mathbf{u} + \beta\mathbf{v} : \alpha, \beta \in \mathbb{R}\}$.

V is at least as large as M .

Is this enough, i.e. is M a linear space?

Yes. Check " $\mathbf{u}, \mathbf{v} \in M \Rightarrow a\mathbf{u} + b\mathbf{v} \in M$ ".

Key Property: LC of LC is LC

$$M \triangleq \{\alpha u + \beta v \mid \alpha, \beta \in \mathbb{R}\}$$

Let's prove:

$$x, y \in M \Rightarrow ax + by \in M$$

$$\begin{aligned} x \in M &\Rightarrow x = \alpha_1 u + \beta_1 v \\ y \in M &\Rightarrow y = \alpha_2 u + \beta_2 v \end{aligned} \Rightarrow ax + by = a(\alpha_1 u + \beta_1 v) + b(\alpha_2 u + \beta_2 v) \\ = [a\alpha_1 + b\alpha_2]u + [a\beta_1 + b\beta_2]v \in M.$$

Key property: LC of LC is LC

Linear combination of two linear combinations of u, v is a linear combination of u, v .

Expanding Any Number of Elements

Eg3: $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is NOT a linear space.

Let's analyze what a minimal space V should contain.

$$M \triangleq \left\{ \alpha_1 \mathbf{u}_1 + \dots + \alpha_n \mathbf{u}_n \mid \alpha_i \in \mathbb{R}, \forall i \right\} \text{ should be in } V.$$

Is this enough? i.e. Is M a linear space?

Yes. Check " $\mathbf{u}, \mathbf{v} \in M \Rightarrow \alpha \mathbf{u} + \beta \mathbf{v} \in M$."

Expanding Any Number of Elements

Eg3: $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is NOT a linear space.

Let's analyze what a minimal space V should contain.

$$M \triangleq \{\alpha_1 \mathbf{u}_1 + \dots + \alpha_n \mathbf{u}_n \mid \alpha_i \in \mathbb{R}, \forall i\}$$

_____ should be in V .

Is this enough? i.e. Is M a linear space?

Yes. Check " $\mathbf{u}, \mathbf{v} \in M \Rightarrow a\mathbf{u} + b\mathbf{v} \in M$."

The collection of **all linear combinations**
of a set of vectors is a linear space.

Span.

Definition: Span

Definition 12 (span)

Suppose V is a linear space.

Suppose $\mathcal{U} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is a subset of V .

The span of $\mathcal{U}$ is defined as

$$\text{span}(\mathcal{U}) \triangleq \{a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k \mid a_1, \dots, a_k \in \mathbb{R}\},$$

In words: the span (of elements of a linear space) is the set of all linear combinations of these elements

Eg: $W = \{s\mathbf{u} + t\mathbf{v} \mid s, t \in \mathbb{R}\}$ is the span of $\{\mathbf{u}, \mathbf{v}\}$.

Remark: For simplicity, we can also say W is the span of $\mathbf{u}, \mathbf{v}$.

Fact: The span of any finite subset of V is a subspace of V .

Analogy: RGB

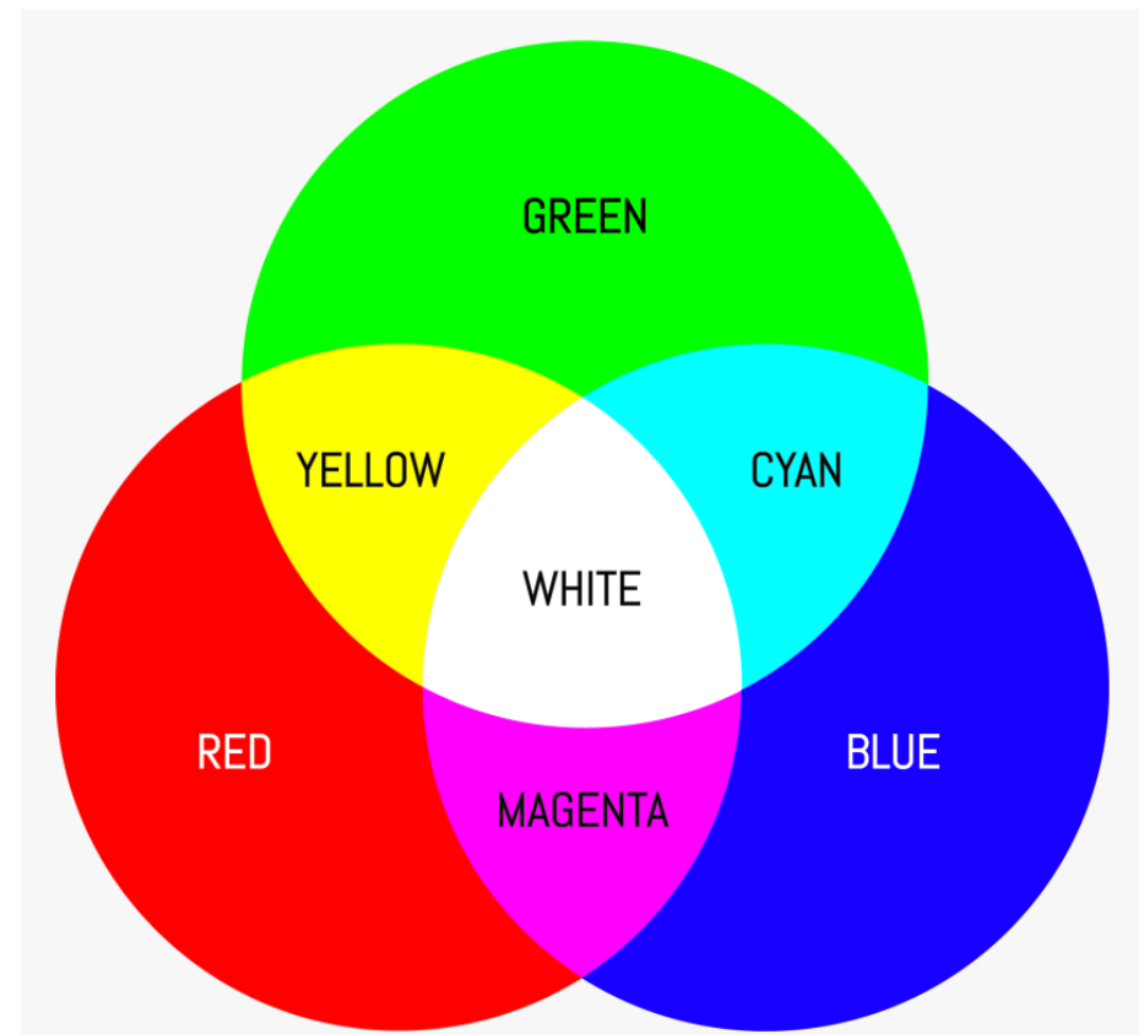
Fact: (Three Prime Colors 三原色)

Any **color** on a computer screen can be represented as a linear combination of Red, Green, and Blue.

Adjusting the intensity of these three colors can produce **any color**.

In LA language:

the set $\{R, G, B\}$ spans the **entire color space**.



Span of Unit Vectors: 2D

Consider two points $e_1=(1,0)$, $e_2=(0,1)$ on the plane.

Prove: $\text{span}\{(1,0), (0,1)\} = \mathbb{R}^2$.

Key Part of the Proof:

$$\text{Any } \begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} \in \text{span}\{(1,0), (0,1)\}$$

A complete proof: next page.

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Proof (in great detail):

Step 0: Analysis of goal.

- 1) To prove two sets $M = H$, often prove $M \subseteq H$ and $H \subseteq M$.
- 2) Since $\mathbb{R}^2$ is a linear space and $e_1, e_2 \in \mathbb{R}^2$, we have $\text{span}(\{e_1, e_2\}) \subseteq \mathbb{R}^2$.
- 3) So, only need to prove $\mathbb{R}^2 \subseteq \text{span}(\{e_1, e_2\})$.

Step 1. Rewrite the goal in math.

Want to prove: Anything in $\mathbb{R}^2$ is also in $\text{span}(\{e_1, e_2\})$,
i.e. $\begin{bmatrix} a \\ b \end{bmatrix} \in \text{span}(\{e_1, e_2\})$ for any $a, b \in \mathbb{R}$

Step 2. Manipulate condition to reach conclusion.

$$\begin{bmatrix} a \\ b \end{bmatrix} = a e_1 + b e_2 \in \text{span}(\{e_1, e_2\}). \text{ Done.}$$

Claim: $\text{span}\{e_1, e_2, \dots, e_n\} = \mathbb{R}^n$.

Reverse: (formal term defined next page)

$\{e_1, e_2, \dots, e_n\}$ is a spanning set of $\mathbb{R}^n$.

Definition: Spanning Set

Definition (spanning set)

Suppose V is a linear space.

Suppose $\mathcal{U} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is a subset of V .

If $\text{span}(\mathcal{U}) = V$, then we say $\mathcal{U}$ is a **spanning set** of V , or **$\mathcal{U}$ spans V** .

Eg: $\{e_1, e_2, \dots, e_n\}$ is a spanning set of $\mathbb{R}^n$.

Eg: $\{\mathbf{u}, \mathbf{v}\}$ is a spanning set of $W = \{s\mathbf{u} + t\mathbf{v} \mid s, t \in \mathbb{R}\}$.

Spanning Set: Example

Exercise: Find a spanning set of $\mathbb{R}^{2 \times 2}$ and justify it.

Remark: Spanning set is NOT unique. Cannot say “**the** spanning set”.

$$E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad E_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad E_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad E_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 2}.$$

$$\text{Any } \begin{bmatrix} a & b \\ c & d \end{bmatrix} = a E_{11} + b E_{12} + c E_{21} + d E_{22} \\ \in \text{span}(\{E_{11}, \dots, E_{22}\})$$

Thus, $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ is a spanning set of $\mathbb{R}^{2 \times 2}$.

Terminology

Equivalent statements:

$\{a_1 \mathbf{u}_1 + \cdots + a_k \mathbf{u}_k \mid a_1, \dots, a_k \in \mathbb{R}\}$ is the span of $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$.

$\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ spans $\{a_1 \mathbf{u}_1 + \cdots + a_k \mathbf{u}_k \mid a_1, \dots, a_k \in \mathbb{R}\}$

$\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is a spanning set of $\{a_1 \mathbf{u}_1 + \cdots + a_k \mathbf{u}_k \mid a_1, \dots, a_k \in \mathbb{R}\}$

Remark: Spanning set is NOT unique. Cannot say “the spanning set”.

Eg: $\{e_1, e_2, \dots, e_n\}$ is a spanning set of $\mathbb{R}^n$.

$\{e_1, e_2, \dots, e_n\}$ spans $\mathbb{R}^n$.

$\mathbb{R}^n$ is the span of $\{e_1, e_2, \dots, e_n\}$.



Summary Today (write Your Own)

One sentence summary:

Detailed summary:

Summary Today (of Instructor)

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We study linear dependence, subspace, span and column space.

Detailed summary:

1. Linear dependence

- Linear dependent elements: trivial linear combination gets 0
- Related to $Ax=0$ having infinitely many solutions

2. Null space

- Solution set of $Ax = 0$ is a linear space (proved)
- Called null space $N(A)$.

3. Span

- Span of a set is the collection of all LC of elements of the set
Property: It is the minimal linear space that contains the set
- Explain why the span is a subspace: LC of LC is LC
- Span of $\{e_i\text{'s}\}$ is $\mathbb{R}^n$.

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